

Using Firedrake to solve a reduced model for the tokamak pedestal density

John Anthony Labbate*

*Columbia Fusion Research Center and
Department of Applied Physics and Applied Mathematics,
Columbia University, New York, New York 10027, USA
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We present a Firedrake-based finite-element implementation of an extended Saarelma–Connor tokamak pedestal electron and neutral-transport model. The extended model is formulated as a coupled system for electron and neutral densities with flexible boundary conditions and initial guesses. Preliminary calculations suggest improved robustness and closer agreement with experimental density profiles than the original model implementation. Among the configurations examined, Dirichlet conditions on the electron density at both radial boundaries, combined with Picard iteration from experimental, piecewise-linear, or hyperbolic-tangent initial profiles yield the best match to the measured pedestal. The legacy solver remains substantially faster than the new implementation under current settings, with the former completing in roughly one quarter of the run time of the latter. Ongoing work will extend systematic comparisons between solvers and broaden validation across experimental datasets. Additionally, timing and free parameter optimization will be explored in future work.

I. INTRODUCTION

Fusion energy is the gold standard of energy production due to its immense fuel efficiency and minimal environmental impact. Magnetic confinement fusion is the leading energy production choice between fusion concepts. A magnetic confinement fusion concept known as the tokamak has shown strong promise for a power plant design due to high confinement operation.

Modern tokamak plasmas operate primarily in two modes - a lower confinement regime (L-mode) and a higher confinement regime (H-mode). H-mode is characterized by the presence of a large transport barrier in the outer few layers of the tokamak plasma. This transport barrier, known as the pedestal, may only traverse a few percent of the total plasma domain [1]. Therefore, the pedestal is characterized by large gradients in both the plasma temperature and density.

The pedestal plays a major role in the plasma performance of a fusion device. For example, the boundary condition imposed by this pedestal sets the temperature profile for the rest of the plasma [2]. However, the pedestal is a major source of instabilities for a tokamak due to the presence of these large gradients. One of these instabilities is called an Edge-Localized Mode (ELM) [1]. ELMs are experimentally observed in most tokamaks, but at a power plant scale, ELMs are expected to be catastrophic for a fusion device [3]. Future fusion devices and power plants will need to operate without ELMs to be successful.

It is clear the tokamak pedestal is of large importance. An understanding of the pedestal could save the tokamak from the problem of ELMs and provide better operating points for devices. However, the pedestal is difficult to model as its large gradients pass through multiple different physics regimes. In 2009, [4] released a model -

EPED - to predict the width and height of the pressure of a tokamak pedestal. EPED combines peeling-ballooning magnetohydrodynamic stability with Kinetic Ballooning Mode (KBM) stability to predict this pressure height and width [4]. While EPED is a very well-studied code (for example, [1]), it has a major drawback of requiring knowledge of the density profile to predict the pressure profile.

In 2023, [2] released a model - the Saarelma-Connor model - that models the density profile of the tokamak pedestal to generate an input for EPED. This is a physics-based model that will be described in detail in Sect. II. Unlike for EPED, a public computational resource for this Saarelma-Connor model does not yet exist. In this study, we develop a computational resource for a modified version of the Saarelma-Connor model. The Saarelma-Connor model makes various assumptions to generate the equation of interest for the electron density in the pedestal. This equation is solved numerically using an implicit method. With an implicit method, it is essential to have a good initial guess, which can be difficult to predict (although [2] offers a tentative solution for this). The need for a good guess for the electron density *a priori* is essential. However, previous work (from [5] has illustrated that this implicit method is difficult to use to find solver convergence and accurate results that best replicate the experimental p-file.

For this reason, we take a novel approach to solving the Saarelma-Connor model. This approach involves stripping lots of the algebra and first integration completed in [2] and beginning with the model's governing equations. We implement a Picard iteration numerical method to solve this system of equations using Firedrake and PETSc. We compare the novel implementation of the Saarelma-Connor model with the original implicit Saarelma-Connor model through this report.

* john.a.labbate@columbia.edu

II. A NEW SOLVER FOR THE SAARELMA-CONNOR MODEL

All variable identities can be found in App. B.

II.1. The original Saarelma-Connor model

The Saarelma-Connor model is a three-fluid model comprised of an electron fluid, a Franck-Condon (FC) monoenergetic neutral fluid, and a charge-exchange (CX) neutral fluid [2]. The governing equations for these fluids in the pedestal region are given in Eq. 13.

$$\nabla \cdot (D_{\text{ped}} \nabla n_e) = -n_e (n_{\text{FC}} + n_{\text{CX}}) S_i, \quad (1)$$

$$\nabla \cdot (V_{\text{FC}} n_{\text{FC}}) = -n_e (n_{\text{FC}} S_i + n_{\text{FC}} S_{\text{CX}}), \quad (2)$$

$$\nabla \cdot (V_{\text{CX}} n_{\text{CX}}) = -n_e (n_{\text{CX}} S_i - \frac{1}{2} n_{\text{FC}} S_{\text{CX}}). \quad (3)$$

Eq. 13 model the diffusive steady-state motion of each fluid with source and sink terms. In Eq. 1 electron impact ionization of both FC and CX neutrals provide an electron source. There are no electron sinks in this model. In Eq. 2, electron impact ionization of FC neu-

trals and charge exchange with a bulk-plasma ion (assuming plasma quasi-neutrality, so the electron and ion densities are locally equal) are both FC neutral sinks. In Eq. 3, electron impact ionization of CX neutrals provide a CX neutral sink, and charge exchange with FC neutrals provides a CX neutral source. The CX neutral source term has a $\frac{1}{2}$ coefficient because it assumes these particles have random trajectories, so half of these neutrals will be ballistically lost to the wall of the device.

In the Saarelma-Connor model, many assumptions regarding the pedestal region are made. Ultimately, the system is reduced to a single 1D ordinary differential equation for the electron density using flux surface averaging, which is defined as $\langle A \rangle = \oint R^2 A d\theta / \oint R^2 d\theta$ for some quantity A . To obtain this 1D equation, an analytic integration is completed. The full derivation of the final equation in the Saarelma-Connor model can be found in [2] and [6], but the final equations are:

$$\frac{d}{dx} (L_2(n_e)) = \frac{n_e S_i}{|V_{\text{FC}}|} \left(1 + \frac{S_{\text{CX}}}{S_i} \right) L_2(n_e), \quad (4)$$

$$\left[1 - \frac{|V_{\text{FC}}| f_{\text{FC}}}{|V_{\text{CX}}| f_{\text{CX}}} \left(\frac{S_i + \frac{S_{\text{CX}}}{2}}{S_i + S_{\text{CX}}} \right) \right] L_2(n_e) = -\frac{C}{|V_{\text{CX}}| f_{\text{CX}}} + \frac{1}{|V_{\text{CX}}| f_{\text{CX}}} D_{\text{ped}} \frac{dn_e}{dx} - \frac{1}{n_e S_i} \frac{d}{dx} \left(\langle |\nabla r|^2 \rangle D_{\text{ped}} \frac{dn_e}{dx} \right). \quad (5)$$

[2] suggests utilizing an implicit solving method due to the stiffness of this third-order equation. The implicit solver works by first solving a simplified model for n_e

where $n_{\text{CX}} = 0$ is assumed (Eq. 6). Then, the full model is solved using Eq. 7 but the n_e within the integral is assumed to be the n_e found in Eq. 6

$$\frac{d}{dx} \left[\langle |\nabla r|^2 \rangle D_{\text{ped}} \frac{dn_e}{dx} \right] = n_e D_{\text{ped}} \frac{S_i + S_{\text{CX}}}{|V_{\text{FC}}|} \left(\frac{dn_e}{dx} - \frac{dn_e}{dx} \Big|_{x=-\infty} \right). \quad (6)$$

$$\begin{aligned} \frac{d}{dx} \left[\langle |\nabla r|^2 \rangle D_{\text{ped}} \frac{dn_e}{dx} \right] = & -n_e S_i \left\{ -\frac{\langle |\nabla r|^2 \rangle D_{\text{ped}}}{|V_{\text{CX}}| f_{\text{CX}}} \left(\frac{dn_e}{dx} - \frac{dn_e}{dx} \Big|_{x=-\infty} \right) \right. \\ & \left. + \left[1 - \frac{|V_{\text{FC}}| f_{\text{FC}} (S_i + S_{\text{CX}}/2)}{|V_{\text{CX}}| f_{\text{CX}} (S_i + S_{\text{CX}})} \right] \langle n_{\text{FC}} \rangle (0) \exp \left[\int_0^x \frac{n_e(x') (S_i + S_{\text{CX}})}{f_{\text{FC}} |V_{\text{FC}}|} dx' \right] \right\}. \end{aligned} \quad (7)$$

Eq. 7 is the form we derived, which includes factors that are missing from [2].

While this solution method may seem advantageous, it relies heavily on the knowledge of the large free parameter space that is included in this model [5]. Specifically, this implicit method has a difficult time converging when implemented with Scipy's `solve.bvp`. This solution method also introduces additional assumptions to the original model, which already is not fully capturing

the behavior of the tokamak pedestal (diffusion equations are really a proxy for more complex transport and turbulence occurring on smaller scales).

II.2. A modified Saarelma-Connor model

In this study, we relax the assumptions imposed by [2] and look for a more cooperative numerical solver by solv-